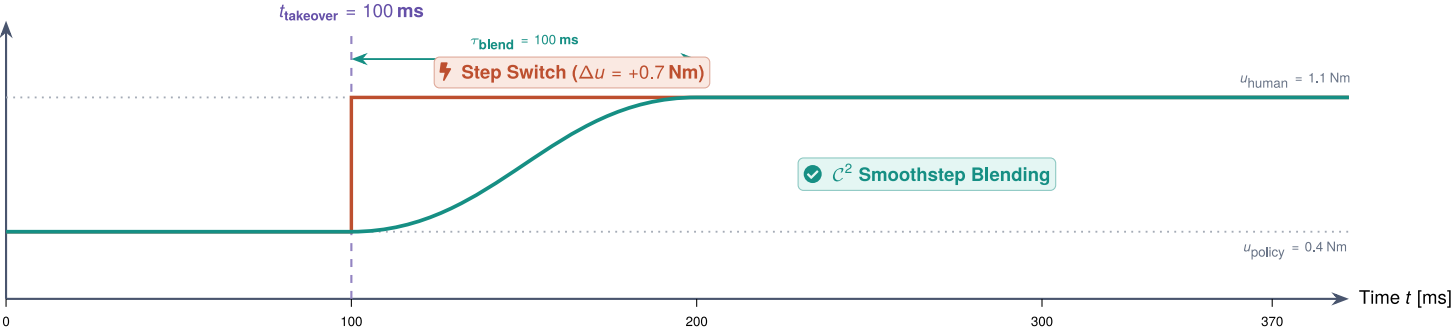


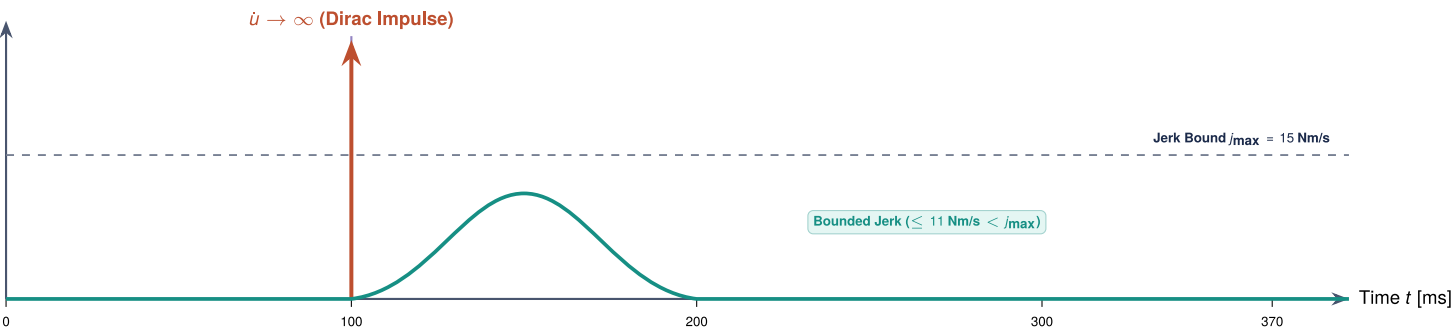
⚡ HARD STEP TAKEOVER VS. C^2 BUMPLESS TORQUE BLENDING

Takeover event at $t = 100\text{ ms}$ with blending transition window $\tau_{\text{blend}} = 100\text{ ms}$

(a) Commanded Control Torque $u(t)$ [Nm]



(b) Torque Rate of Change $\dot{u}(t) \propto \ddot{q}(t)$ (Actuator Jerk) [Nm/s]



(c) Actuator Phase Current $i(t)$ [A] & Gearbox Backlash Stress

